

ABM BUSINESS LOGIC BIBLE

DEEP COHERENCE AUDIT

CONSOLIDATED MASTER DOCUMENT

Cross-Section Conflict Analysis Across All 19 Sections

V1.3 FSD + V2 Deep Specification | Decimal Technologies

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Classification: INTERNAL | Engineering-Ready Specification

This audit systematically cross-references every rule, threshold, timing window, state transition, decision gate, and architectural constraint across all 19 sections of the ABM Business Logic Bible (V1.3 FSD base document + V2 deep-specification layers built across Sections 1-16).

Every conflict identified traces to a specific engine failure mode: deadlock, contradictory automation, data corruption, capital waste, or governance collapse. Conflicts are classified by severity:

CRITICAL - Engine will produce incorrect behavior, deadlock, or data corruption. Must resolve before implementation.

HIGH - Engine will produce suboptimal behavior, wasted capital, or governance gaps. Must resolve before production.

MEDIUM - Engine will function but with ambiguity that creates implementation risk. Resolve during engineering review.

Audit Method: Each section's rules are cross-referenced against every other section. Conflicts are grouped by architectural domain, not by section number, because the most dangerous conflicts span multiple sections. The Master Resolution Priority table orders conflicts by engine-collapse risk.

AUDIT FINDINGS SUMMARY

Total Conflicts Identified: 34

CRITICAL (engine-breaking): 11

HIGH (governance/capital risk): 14

MEDIUM (ambiguity/implementation risk): 9

Conflict Categories:

A. Anchor Model Conflicts (Account vs Opportunity) - 5 conflicts

B. Scoring and Tier Conflicts - 6 conflicts

C. State Machine and Lifecycle Conflicts - 5 conflicts

D. AI Agent and Autonomy Conflicts - 4 conflicts

E. Timing and Sequence Conflicts - 4 conflicts

F. Integration and Data Flow Conflicts - 4 conflicts

G. Governance and Learning Loop Conflicts - 3 conflicts

H. KSA-Specific and Compliance Conflicts - 3 conflicts

THREE ENGINE-COLLAPSING CONFLICT CLUSTERS

Cluster 1 - The Anchor Model Crisis (COH-A01 through A05)

Every section downstream of Section 1 still assumes Account is the primary entity, but V2 re-anchored to Opportunity (E38). The scoring engine scores Accounts. The state machine tracks Account states. The CRM syncs Accounts. But decisions should be made on Opportunities. This single unresolved tension ripples through scoring, tiers, enrichment, outreach, engagement tracking, and CRM sync.

Cluster 2 - The Dual Scoring Crisis (COH-B01)

Two completely different scoring equations exist: V1.3's four-dimension weighted composite (0-100 scalar) and V2 Section 1's multiplicative Effective Opportunity chain (unbounded output). The 75/50 tier thresholds are calibrated for the first equation. No specification exists for which the engineer should implement.

Cluster 3 - State-Tier Confusion (COH-C01, C02)

The word 'HOT' means two completely different things in the same document (a lifecycle state AND a scoring tier), and there is no specified behavior for what happens when an Account's tier changes while a sequence is mid-execution.

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KEY ARCHITECTURAL CONTEXT

Before reading individual conflicts, the reader must understand the foundational architecture that these conflicts span. This section provides the minimum context needed to evaluate each conflict.

Architectural Element	Specification	Source
Unit of Selling	Opportunity (Entity 38 / E38). Account is the relationship contained in	V2 Section 1
Decision Architecture	25 Meta-Principles > ~230 Operational Rules > 8-Step Universal Decision Sequence	V2 Section 1
Universal Decision Sequence	Legal > Safety > Relationship > Readiness > Timing > Capital > V2 Section 1	V2 Section 1
Autonomy Ladder	A0 (Hard Stop) through A5 (Full Autonomy). A0 = structurally imposed	V2 Section 5
Scoring Equation (V1.3)	Weighted composite: Signal(35%) + Regulatory(30%) + Reachability(20%) + Warmth(15%)	V2 Section 1
Scoring Equation (V2)	Effective Opp = Dynamic Score x ICS/100 x Stage x Budget x Environmental x Risk x Window	V2 Section 1
Tier Thresholds	HOT >= 75, WARM >= 50, COLD < 50 (calibrated for V1.3 0-100 scale)	V2 Section 4.4
Entity Model	38 entities across 7 tiers, 20 modules (M0-M19) + orchestration	V2 Section 2-3
Integration Spine	EPIS Reality Confidence Model governs all integrations	V2 Section 4
Constraint Hierarchy	6 tiers: Legal/Compliance > Safety > Relationship > Timing > Readiness > Optimization	V2 Section 1
Constitutional Constraint	Section 13 establishes Section 1 Meta-Principles as non-negotiable (learning loop overrides)	V2 Section 13
Capital Model	5 types: Attention, Relationship, Connector, Brand, System. EROV2 Section 1	V2 Section 1
Orchestration Constraint	n8n holds zero business logic (ORCH-NOLOGIC-001)	V2 Section 3

CATEGORY A: ANCHOR MODEL CONFLICTS

The single most architecturally dangerous category. V2 Section 1 re-anchored the unit of selling from Account to Opportunity (Entity 38). But V1.3's entire pipeline, database schema (Section 19), state machine (Section 23), scoring engine (Section 4), and CRM sync (Section 9/12) all assume Account is the primary entity. Every downstream section inherits this contradiction unless explicitly resolved.

[CRITICAL] COH-A01: Account-Centric vs Opportunity-Centric Design

Sections: V2 Section 1 (Meta-Principles) vs V1.3 Principle 2 vs V1.3 Section 19 (DB Schema) vs V1.3 Section 23 (State Machine)

Conflict Description: V2 Section 1 establishes Opportunity (E38) as the unit of selling with Account as the relationship container. V1.3 Principle 2 explicitly states the platform is account-centric with all workflows revolving around ACCOUNT. V1.3 Section 19 Entity Relationship Model shows ACCOUNT as the root with all other entities as children. V1.3 Section 23 defines account states (NEW, ENRICHED, OUTREACH_ACTIVE, REPLIED, OPPORTUNITY, DORMANT) — all scoped to Account, not Opportunity.

Rule A: V2 Section 1: Opportunity (E38) is the unit of selling. Account is the relationship container. Decision Score = $f(\text{Effective_Opportunity})$.

Rule B: V1.3 Principle 2 + Section 19 + Section 23: Account is the root entity. All states, workflows, and scoring are Account-scoped.

Engine Impact: The engineer cannot implement both. If they follow V1.3, the Opportunity re-anchoring is dead code. If they follow V2, the entire state machine, scoring engine, and CRM sync need rebuilding.

Recommended Resolution: RESOLVE BY: V2 governs. Account is the relationship container; Opportunity is the decision unit. Add ANCHOR-001: 'Account holds relationships and firmographic data. Opportunity holds deal-specific state, scoring, and workflow. All execution decisions (enrichment, outreach, handoff) are Opportunity-scoped. All relationship decisions (tone, channel preference, contact history) are Account-scoped.' Retrofit V1.3 Sections 19 and 23 to support dual-entity model.

[CRITICAL] COH-A02: Score Entity Anchor: Account Score vs Opportunity Score

Sections: V1.3 Section 4 (Scoring Engine) vs V2 Section 1 (Effective Opportunity Equation)

Conflict Description: V1.3 Section 4 computes a composite score (0-100) per Account using four weighted dimensions: Signal Strength (35%), Regulatory Pressure (30%), Reachability (20%), Warmth (15%). V2 Section 1 defines Effective Opportunity as a multiplicative chain: $\text{Dynamic Score} \times \text{ICS}/100 \times \text{Stage Bonus} \times \text{Budget Modifier} \times \text{Entrenchment Modifier} \times \text{Risk Modifier} \times \text{Window Multiplier}$. The V1.3 score is Account-level. The V2 Effective Opportunity is Opportunity-level with modifiers that are Opportunity-specific (Stage Bonus, Window Multiplier).

Rule A: V1.3 Section 4: Composite score per Account. 0-100 scale.

Rule B: V2 Section 1: Effective Opportunity per Opportunity. Unbounded multiplicative chain. Decision Score = $\text{Effective Opportunity} \times \text{Strategic Readiness} / 100 / \text{Capital Cost}$.

Engine Impact: If the score is Account-level but decisions are Opportunity-level, the engine cannot correctly route two Opportunities from the same Account at different stages. The two scoring models are incompatible at the entity level.

Recommended Resolution: RESOLVE BY: The composite score is Account-level (measures account attractiveness). Effective Opportunity is computed per-Opportunity, inheriting the Account's composite score plus Opportunity-specific modifiers. The Section 4 score is an INPUT to the Section 1 equation, not a replacement. Both coexist. Section 4 score triggers tier assignment; Effective Opportunity governs resource allocation within a tier.

[CRITICAL] COH-A03: Tier Assignment: Account Tier vs Opportunity Tier

Sections: V2 Section 5 (Tier Assignment) vs V2 Section 1 (Opportunity Anchor)

Conflict Description: V2 Section 5 specifies HOT/WARM/COLD tiers assigned to ACCOUNTS based on the composite score. But if an Account has two Opportunities — one HOT, one COLD — what is the Account's tier? The tier gates enrichment and outreach. If the Account is rated by its best Opportunity, COLD Opportunities on HOT Accounts get free enrichment. If rated by its worst, HOT Opportunities on COLD Accounts get blocked.

Rule A: V2 Section 5: Tier assigned per Account. HOT $\geq$ 75 triggers enrichment + outreach.

Rule B: V2 Section 1: Opportunity is the decision unit. Multiple Opportunities per Account possible.

Engine Impact: Engine deadlock: An Account with one active Opportunity in OUTREACH_ACTIVE and a new signal suggesting a second Opportunity cannot be correctly routed.

Recommended Resolution: RESOLVE BY: Account tier = MAX(Opportunity tiers) for relationship-level decisions. Opportunity tier = its own Effective Opportunity score for execution decisions. Add TIER-MULTI-001: 'When an Account has multiple Opportunities, the Account tier reflects the highest-tier Opportunity. Workflow routing operates on Opportunity tier, not Account tier.'

[HIGH] COH-A04: CRM Sync: HubSpot Deal vs Account Object Mapping

Sections: V1.3 Section 9 (Human Handoff) vs V1.3 Section 12 (CRM Sync) vs V2 Section 12

Conflict Description: V1.3 Section 9 specifies 'Create CRM Opportunity' as a handoff action. V1.3 Section 12 lists POST /sync-hubspot to synchronize opportunities and engagement. But the internal Opportunity (E38) is not the same as a HubSpot Deal. The internal Opportunity exists from the moment the engine identifies a selling chance; the HubSpot Deal should only be created when a human handoff occurs. No rule specifies when E38 is created vs when HubSpot Deal is created.

Rule A: V1.3 Section 9: CRM Opportunity = handoff trigger. Created when human takes over.

Rule B: V2 Section 1: Opportunity (E38) = unit of selling. Created when engine identifies opportunity.

Engine Impact: If E38 creation triggers HubSpot Deal creation, the CRM floods with unqualified deals. If only handoff creates the Deal, internal reporting loses Opportunity tracking.

Recommended Resolution: RESOLVE BY: E38 created internally when engine identifies opportunity (early). HubSpot Deal created only at handoff stage. Add CRM-OPP-001: 'Internal Opportunity (E38) and CRM Deal are separate lifecycle objects. E38 exists from signal-to-opportunity conversion. HubSpot Deal exists from human handoff. The sync maps E38 to Deal only after handoff gate passes.'

[HIGH] COH-A05: Engagement Events: Account vs Opportunity Attribution

Sections: V1.3 Section 8 (Engagement Tracking) vs V2 Section 1 (Opportunity Anchor)

Conflict Description: V1.3 Section 8 tracks engagement events (email opens, clicks, replies) at the Account/Contact level. V2 Section 1 says the Opportunity is the decision unit. An engagement event from a contact at a multi-Opportunity Account must be attributed to the correct Opportunity for scoring to work. But V1.3's engagement tracking has no Opportunity field — events are linked to Account + Contact.

Rule A: V1.3 Section 8: Engagement events linked to Account + Contact only.

Rule B: V2 Section 1: Opportunity-level decisions require Opportunity-level engagement data.

Engine Impact: Engagement from a contact discussing Opportunity A gets attributed to Opportunity B (or to the Account generically), corrupting Effective Opportunity scores for both.

Recommended Resolution: RESOLVE BY: Add opportunity_id to engagement events. Add ENG-OPP-001: 'Each outreach message is tagged with the Opportunity ID it was sent about. Engagement events (opens, clicks, replies) inherit the Opportunity ID from the triggering message. If a contact engages with a message not tagged to an Opportunity, the event is attributed to the Account and flagged for manual routing.'

CATEGORY B: SCORING AND TIER CONFLICTS

The scoring engine gates every downstream decision: enrichment, outreach, resource allocation, and handoff timing. Conflicts in scoring produce cascading failures across the entire pipeline.

[CRITICAL] COH-B01: Two Competing Scoring Models: V1.3 Composite vs V2 Effective Opportunity

Sections: V1.3 Section 4 vs V2 Section 1 (Meta-Principles) vs V2 Section 5 (Tier Assignment)

Conflict Description: V1.3 Section 4 defines a four-dimension weighted composite (Signal 35%, Regulatory 30%, Reachability 20%, Warmth 15%) producing a 0-100 scalar. V2 Section 1 defines Effective Opportunity as a multiplicative chain with unbounded output. The tier thresholds (HOT >= 75, WARM >= 50) are calibrated for the 0-100 scale. If the V2 equation is used, these thresholds are meaningless because the output range is different.

Rule A: V1.3 Section 4: Composite = 0.35*Signal + 0.30*Regulatory + 0.20*Reachability + 0.15*Warmth. Output: 0-100.

Rule B: V2 Section 1: Effective Opportunity = DynamicScore x ICS/100 x Stage x Budget x Entrenchment x Risk x Window. Output: unbounded.

Engine Impact: The engineer cannot apply 75/50 thresholds to an unbounded multiplicative chain. The entire tier system, which gates enrichment and outreach, collapses.

Recommended Resolution: RESOLVE BY: Dual-score architecture. V1.3 composite is the Account Attractiveness Score (0-100, gates tier). V2 Effective Opportunity is computed per-Opportunity, used for prioritization within a tier and resource allocation. Add SCORE-DUAL-001: 'Tier assignment uses Account Attractiveness Score with existing 75/50 thresholds. Effective Opportunity score governs Opportunity-level prioritization and capital allocation.'

[HIGH] COH-B02: Hysteresis Band vs Weight Recalibration Interaction

Sections: V2 Section 5 (Tier Hysteresis) vs V2 Section 10 (Analytics/Learning) vs V1.3 Section 10.3

Conflict Description: V2 Section 5 specifies a hysteresis band (e.g., +/- 5 points) to prevent tier oscillation. V1.3 Section 10.3 says the analytics engine should 'Update Scoring Weights' based on performance data. If scoring weights change, previously stable accounts near the threshold could flip tiers despite the hysteresis band, because the underlying score changed, not just the threshold.

Rule A: V2 Section 5: Hysteresis band prevents tier flip within +/- 5 points of threshold.

Rule B: V1.3 Section 10.3: Analytics engine updates scoring weights automatically.

Engine Impact: Weight recalibration that shifts all scores by 3 points bypasses hysteresis silently. The band prevents flipping on single-signal noise but not on systemic recalibration drift.

Recommended Resolution: RESOLVE BY: Add HYST-RECAL-001: 'When scoring weights are recalibrated, the system first computes the shift in mean score across all accounts. If mean shift > 2 points, the hysteresis band widens proportionally for 72 hours (transition period). Any tier changes triggered by recalibration alone (not by new signals) are flagged for human review before execution.'

[HIGH] COH-B03: Three Different Freshness/Decay Models

Sections: V1.3 Section 3 (Signal) vs V1.3 Section 19 (DB) vs V2 Section 4 (EPIS) vs V2 Section 16 (Temporal)

Conflict Description: V1.3 Section 3 uses a 'freshness window' for duplicate detection. V1.3 Section 19 stores signal 'freshness' as a field. V2 Section 4 (EPIS) has a time-decay function on confidence scores. V2 Section 16 (Temporal Intelligence Layer) defines temporal decay curves for different signal types. These four freshness/decay models are not reconciled — a signal could be 'fresh' for duplicate detection but 'decayed' for scoring, or vice versa.

Rule A: V1.3 Section 3: Freshness window for deduplication (binary: fresh or stale).

Rule B: V2 Section 4 + Section 16: Continuous decay functions (EPIS confidence degrades over time; temporal curves vary by signal type).

Engine Impact: A signal marked 'fresh' (not duplicate) but with high temporal decay gets scored at near-zero impact. A 'stale' signal that just expired its freshness window but was highly impactful yesterday disappears from scoring entirely.

Recommended Resolution: RESOLVE BY: Unify under a single freshness model. Add FRESH-UNIFY-001: 'Signal freshness is a continuous 0-1 value computed by the temporal decay curve (Section 16). Duplicate detection uses freshness > 0.8 (not binary). EPIS confidence incorporates freshness as a multiplicative modifier. Section 16 temporal curves are the authoritative decay model.'

[HIGH] COH-B04: Scoring Frequency and Debounce: When Does Rescoring Happen?

Sections: V1.3 Section 4.3 (Scoring Workflow) vs V1.3 Section 3 (Signal Triggers) vs V2 Section 15 (Resilience)

Conflict Description: V1.3 Section 3 says every stored signal triggers scoring. V1.3 Section 4.3 shows scoring as event-driven (new signal received -> compute score). If 10 signals arrive in 5 minutes (e.g., a news burst), the scoring engine runs 10 times. V2 Section 15 (Resilience) specifies debounce mechanisms for system stability but does not specify scoring-specific debounce windows.

Rule A: V1.3 Section 3 + 4.3: Every signal triggers immediate rescoring.

Rule B: V2 Section 15: System-level debounce for stability (no scoring-specific parameter).

Engine Impact: 10 signals in 5 minutes = 10 rescoring runs, 10 potential tier flips, 10 potential enrichment triggers. Each enrichment trigger consumes API credits. The engine burns capital on noise bursts.

Recommended Resolution: RESOLVE BY: Add SCORE-DEBOUNCE-001: 'Scoring is debounced with a 5-minute coalescing window. Multiple signals arriving within the window are batched. Rescoring runs once after the window closes, incorporating all new signals. Emergency override: if any signal in the batch is Priority CRITICAL, scoring runs immediately.'

[MEDIUM] COH-B05: Static Signal Priority Levels vs Contextual Significance

Sections: V1.3 Section 3 (SIG-004: Priority Levels) vs V2 Section 1 (Contextual Scoring)

Conflict Description: V1.3 SIG-004 assigns static priority levels to signals. V2 Section 1 emphasizes contextual scoring — the same signal type may have different significance depending on the account's current state, relationship history, and timing. A SAMA regulatory announcement is high-priority generically, but may be irrelevant for a specific bank already compliant.

Rule A: V1.3 SIG-004: Static priority levels per signal type.

Rule B: V2 Section 1: Contextual significance varies by account state, history, and timing.

Engine Impact: Static priorities waste attention on generically important but contextually irrelevant signals. Engineering may implement static priorities (simpler) and never add contextual modifiers.

Recommended Resolution: RESOLVE BY: Add SIG-CONTEXT-001: 'Signal priority has two components: base priority (static, per signal type) and contextual modifier (dynamic, per account). Final priority = base x contextual. Contextual modifier considers: account tier, current state, signal relevance to known opportunities, and relationship stage. MVP: base priority only. P2: contextual modifiers.'

[MEDIUM] COH-B06: Weight Floor Constraints vs Learning Loop Freedom

Sections: V2 Section 1 (Meta-Principles) vs V2 Section 10 (Reality Calibration) vs V2 Section 13 (Learning Governance)

Conflict Description: V2 Section 1 defines scoring dimensions with specific weights (e.g., Signal 35%). V2 Section 10 allows the analytics engine to recalibrate weights based on performance data. V2 Section 13 establishes constitutional constraints that prevent learning loops from overriding meta-principles. But nowhere is a weight floor specified — can the learning loop reduce Signal Strength weight to 0%?

Rule A: V2 Section 1: Initial weights defined (Signal 35%, Regulatory 30%, etc.).

Rule B: V2 Section 13: Learning loops cannot override meta-principles but no weight-specific constraint.

Engine Impact: A learning loop could theoretically zero out a dimension, making the scoring engine ignore an entire class of signals. This is technically not a meta-principle violation but defeats the scoring design.

Recommended Resolution: RESOLVE BY: Add WEIGHT-FLOOR-001: 'No scoring dimension weight may be reduced below 10% or increased above 50% by any learning loop. Weight changes > 5 percentage points require human approval. These floors are constitutional constraints per Section 13.'

CATEGORY C: STATE MACHINE AND LIFECYCLE CONFLICTS

The state machine governs what the engine can and cannot do at any given moment. State conflicts produce the most visible operational failures: duplicate outreach, dropped sequences, inconsistent CRM records.

[CRITICAL] COH-C01: HOT State vs HOT Tier: Name Collision

Sections: V1.3 Section 23.2 (Account States) vs V1.3 Section 4.4 (Tier Decision Logic)

Conflict Description: V1.3 Section 23.2 does not explicitly name a state 'HOT' but the enrichment workflow in Section 5 starts with 'HOT Account Detected.' V1.3 Section 4.4 defines HOT as a scoring tier (≥ 75). The word 'HOT' is used interchangeably for both a lifecycle trigger and a scoring classification. An Account can be HOT-tier (score ≥ 75) but in state ENRICHED (already processed). Or it could be in enrichment (state) but score-wise WARM (tier drop during enrichment).

Rule A: V1.3 Section 5: 'HOT Account Detected' triggers enrichment (lifecycle event).

Rule B: V1.3 Section 4.4: HOT = score ≥ 75 (scoring classification).

Engine Impact: Engineers will conflate tier and state. A tier downgrade during enrichment could halt enrichment mid-process. Or a tier upgrade could trigger duplicate enrichment for an already-enriched account.

Recommended Resolution: RESOLVE BY: Rename scoring tiers to TIER_1 (was HOT), TIER_2 (was WARM), TIER_3 (was COLD). Reserve lifecycle language for states. Add TIER-NAME-001: 'Scoring tiers use numbered labels (TIER_1/2/3). Account states use lifecycle labels (NEW, ENRICHED, OUTREACH_ACTIVE, etc.). No state name may duplicate a tier name.'

[CRITICAL] COH-C02: Mid-Outreach Tier Downgrade: No Transition Defined

Sections: V1.3 Section 7 (Outreach Sequences) vs V1.3 Section 4 (Scoring) vs V2 Section 8 (Trust Preservation)

Conflict Description: V1.3 Section 7 launches outreach sequences for HOT-tier accounts. V1.3 Section 4 rescores on every new signal. If a signal causes a tier downgrade from HOT to WARM while a sequence is mid-execution (e.g., email 3 of 5 sent), no rule specifies what happens. Options: abort sequence (wastes emails 1-2), continue sequence (sends HOT-tier messages to a WARM account), or pause and wait for re-qualification.

Rule A: V1.3 Section 7: Outreach sequence continues until completion or reply.

Rule B: V1.3 Section 4: Rescoring can change tier at any time.

Engine Impact: An account receiving a HOT-tier aggressive sequence after downgrading to WARM gets mismatched messaging intensity. V2 Section 8 (Trust Preservation) would classify this as a trust violation — over-engagement destroys relationship capital.

Recommended Resolution: RESOLVE BY: Add STATE-TIER-MID-001: 'If an Account downgrades from TIER_1 to TIER_2 during an active outreach sequence: (a) Pause the current sequence immediately. (b) Re-evaluate the Opportunity's Effective Score. (c) If the Opportunity is still active, transition to a TIER_2 sequence at the equivalent step. (d) If the Opportunity is dead, terminate the sequence and log the reason. Upgrade during sequence: continue current sequence (do not escalate mid-sequence).'

[HIGH] COH-C03: Reply Detection: Unconditional Handoff vs Intent Classification

Sections: V1.3 Section 8 (Engagement) vs V1.3 Section 9 (Handoff) vs V2 Section 7 (Trust Construction)

Conflict Description: V1.3 Section 8 says positive reply triggers handoff. V1.3 Section 9 lists handoff criteria including 'Reply detected.' But V2 Section 7 (Trust Construction) recognizes that not all replies indicate buying intent — some are out-of-office, some are 'unsubscribe,' some are 'not interested.' V1.3 treats any reply as a handoff signal; V2 requires intent classification before handoff.

Rule A: V1.3 Section 8-9: Reply detected -> pause automation -> notify sales -> handoff.

Rule B: V2 Section 7: Intent must be classified before handoff. Not all replies are buying signals.

Engine Impact: Out-of-office replies trigger handoff, wasting sales team attention. 'Not interested' replies trigger handoff instead of routing to a nurture path or dormancy.

Recommended Resolution: RESOLVE BY: Add REPLY-INTENT-001: 'All replies are first classified by the AI Intent Agent into: POSITIVE (buying signal), NEUTRAL (information request), NEGATIVE (rejection), OOO (auto-reply), UNSUBSCRIBE (opt-out). Only POSITIVE triggers handoff. NEUTRAL continues sequence with adjusted tone. NEGATIVE routes to nurture with 90-day cooldown. OOO reschedules. UNSUBSCRIBE is a hard stop.'

[HIGH] COH-C04: DORMANT State Re-activation Path Missing

Sections: V1.3 Section 23.2 (States) vs V2 Section 16 (Temporal Intelligence) vs V2 Section 8 (Trust Preservation)

Conflict Description: V1.3 Section 23.2 lists DORMANT as a terminal state after 'Extended inactivity.' V2 Section 16 (Temporal Intelligence) implies that dormant accounts should be periodically re-evaluated as market conditions change. V2 Section 8 (Trust Preservation) specifies cooldown periods. But no rule defines: how long is 'extended inactivity'? What triggers DORMANT -> re-activation? Does re-activation require a new signal or can it be time-triggered?

Rule A: V1.3 Section 23.2: DORMANT = terminal state. No exit transition defined.

Rule B: V2 Sections 8 + 16: Temporal decay and trust recovery imply re-activation should be possible.

Engine Impact: DORMANT accounts are permanently lost to the engine even if new high-value signals appear. The engine accumulates dead weight over time, shrinking the active portfolio.

Recommended Resolution: RESOLVE BY: Add DORMANT-REACT-001: 'DORMANT is not terminal. Re-activation triggers: (a) New signal with priority >= HIGH for the account. (b) Quarterly review cycle (Section 16 temporal trigger). (c) Manual override by sales team. Re-activated accounts receive a DORMANT_HISTORY tag that informs outreach tone. Minimum dormancy period: 90 days (trust preservation).'

[MEDIUM] COH-C05: Account State OPPORTUNITY vs Opportunity Entity: Circular Definition

Sections: V1.3 Section 23.2 (Account State 'OPPORTUNITY') vs V2 Section 1 (Entity 38 'Opportunity')

Conflict Description: V1.3 Section 23.2 has an account state called 'OPPORTUNITY' which means CRM opportunity created. V2 Section 1 has an entity called 'Opportunity' (E38) which is the unit of selling. The state is a lifecycle marker; the entity is a data object. An Account can be in state OPPORTUNITY without an E38 entity existing (if CRM opportunity was created manually), and an E38 can exist before the Account reaches state OPPORTUNITY.

Rule A: V1.3 Section 23.2: Account state 'OPPORTUNITY' (lifecycle marker).

Rule B: V2 Section 1: Entity 38 'Opportunity' (data object, unit of selling).

Engine Impact: State and entity are coupled by name but not by lifecycle. Engineers will assume they are synchronized, creating bugs where the state changes but the entity does not exist or vice versa.

Recommended Resolution: RESOLVE BY: Rename account state 'OPPORTUNITY' to 'DEAL_ACTIVE' to disambiguate. Add invariant: 'Account state DEAL_ACTIVE requires at least one Opportunity entity in status OPEN or NEGOTIATING. An Account cannot enter DEAL_ACTIVE without a corresponding E38 record.'

CATEGORY D: AI AGENT AND AUTONOMY CONFLICTS

The AI layer sits between intelligence and execution. Conflicts between the multi-agent architecture, the Autonomy Ladder, and the human-in-the-loop requirements create the most dangerous category of silent failures.

[CRITICAL] COH-D01: MVP 'Review All AI Output' vs Autonomy Ladder A2 (Autonomous with Audit)

Sections: V1.3 Section 16.4 (Operational Principles) vs V2 Section 5 GOV (Autonomy Ladder) vs V1.3 Section 15.7

Conflict Description: V1.3 Section 16.4 says 'Review all AI output initially.' V1.3 Section 15.7 says 'Sending without human review = Enterprise brand risk. Manual approval during MVP.' But V2's Autonomy Ladder defines A2 as 'Autonomous with Audit' — the system acts and logs for review, not requiring pre-approval. Several module operations in V2 are classified as A2, meaning they would execute without human pre-approval, directly contradicting the MVP constraint.

Rule A: V1.3 Sections 15.7 + 16.4: All AI output requires human review during MVP.

Rule B: V2 Autonomy Ladder: A2 operations execute first, review after. Multiple modules classified A2.

Engine Impact: The engineer either implements A2 (violating MVP safety) or forces A1 everywhere (defeating the Autonomy Ladder design). Worse: if different engineers implement different interpretations, some modules will be autonomous while others are gated, creating inconsistent behavior.

Recommended Resolution: RESOLVE BY: Add AUTO-MVP-001: 'During MVP phase (Phase 1), all operations cap at A1 (Recommend and Wait) regardless of their designed autonomy level. The Autonomy Ladder defines TARGET autonomy, not initial autonomy. Promotion from A1 to A2+ requires: (a) Minimum 100 decisions at current level with < 5% override rate. (b) Human governance approval. (c) 30-day shadow period at new level.'

[HIGH] COH-D02: QC Agent Temperature vs Generation Agent Temperature Retry Loop

Sections: V1.3 Section 15.6 (Prompt Pitfalls) vs V1.3 Section 7 (AI Architecture) vs V2 Section 11 (Decision Partnership)

Conflict Description: V1.3 Section 15.6 warns against incorrect temperature settings. V1.3 Section 7 specifies the AI Research Agent and AI Writer Agent as separate agents. The QC Agent evaluates output quality. If QC fails, the system regenerates. But if the Generation Agent uses temperature 0.7 (creative) and the QC Agent evaluates at temperature 0.0 (deterministic), the QC Agent may consistently reject outputs that are within acceptable quality range, creating an infinite regeneration loop.

Rule A: V1.3 Section 7: Generation Agent (creative temperature) -> QC Agent (strict evaluation).

Rule B: V1.3 Section 15.6: Temperature misconfiguration is a named pitfall but no resolution specified.

Engine Impact: Infinite regeneration loop burns API credits and delays outreach. The QC Agent becomes a bottleneck that blocks all AI-generated content without producing useful feedback.

Recommended Resolution: RESOLVE BY: Add QC-TEMP-001: 'QC Agent uses a rubric-based evaluation, not regeneration of the same output. QC temperature is 0.0 (deterministic scoring). Maximum regeneration attempts: 3. After 3 failures, escalate to human review with QC failure reasons. QC rubric includes: factual accuracy, tone match, personalization depth, compliance, and length.'

[HIGH] COH-D03: n8n Zero Business Logic vs Workflow Decision Nodes

Sections: V2 Section 3 (ORCH-NOLOGIC-001) vs V1.3 Section 21 (n8n Architecture)

Conflict Description: V2 Section 3 establishes ORCH-NOLOGIC-001: n8n holds zero business logic. V1.3 Section 21 describes n8n as the orchestration layer. But n8n workflows inherently contain IF/ELSE nodes, switch nodes, and conditional routing. The boundary between 'orchestration routing' and 'business logic' is not defined. Is 'if account tier == HOT, route to enrichment' business logic or routing?

Rule A: V2 Section 3: ORCH-NOLOGIC-001: n8n contains zero business logic.

Rule B: V1.3 Section 21: n8n contains workflow routing with conditional nodes.

Engine Impact: Without a clear boundary, engineers will put business logic in n8n nodes because it is convenient. Once business logic lives in n8n, the rule audit system cannot reach it, creating ungoverned rules.

Recommended Resolution: RESOLVE BY: Add ORCH-BOUNDARY-001: 'n8n may contain: (a) event routing (signal -> correct handler), (b) data transformation (format conversion), (c) error handling (retry, dead letter). n8n may NOT contain: (a) threshold comparisons (score >= 75), (b) state transitions, (c) scoring calculations, (d) any if/else that references business entities or their attributes. All business decisions are API calls to the FastAPI backend. n8n routes events; FastAPI makes decisions.'

[MEDIUM] COH-D04: Hallucination Prohibition vs Hypothesis Generation

Sections: V1.3 Section 16.3 (AI Principles) vs V2 Section 1 (Strategic Reasoning) vs V2 Section 6 (Knowledge Acquisition)

Conflict Description: V1.3 Section 16.3 specifies 'Low hallucination tolerance.' V1.3 Section 15.5 warns against hallucinated statistics. But V2 Section 1 (Strategic Reasoning as Core Intelligence) and V2 Section 6 (Knowledge Acquisition Engine) require the AI to generate hypotheses about account needs, predict buying intent, and reason about unobserved states. Hypothesis generation is inherently speculative — the line between a hypothesis and a hallucination is context-dependent.

Rule A: V1.3 Section 16.3 + 15.5: Zero tolerance for hallucination in outputs.

Rule B: V2 Sections 1 + 6: AI must generate hypotheses and reason about unobserved states.

Engine Impact: If hallucination prohibition is absolute, the AI cannot generate hypotheses (all hypotheses are technically unverified). If hypotheses are allowed, where is the boundary?

Recommended Resolution: RESOLVE BY: Add HYPOTHESIS-001: 'Distinguish between OUTPUT hallucination (fabricating facts in customer-facing messages — prohibited) and INTERNAL hypothesis (reasoning about probabilities in the intelligence layer — required). Internal hypotheses must carry confidence scores and must never surface in customer-facing content unless verified by a signal with EPIS >= 0.7. The AI may hypothesize internally; it may not hallucinate externally.'

CATEGORY E: TIMING AND SEQUENCE CONFLICTS

[HIGH] COH-E01: Email and LinkedIn Channel Collision: No Coordination Rule

Sections: V1.3 Section 7 (Outreach Sequences) vs V2 Section 7 (Trust Construction) vs V2 Section 1 (Capital Model)

Conflict Description: V1.3 Section 7 specifies both email and LinkedIn as outreach channels. V2 Section 7 (Trust Construction) treats each touchpoint as a capital expenditure. V2 Section 1's capital model tracks attention capital. But no rule specifies the minimum gap between an email and a LinkedIn message to the same contact. Without coordination, the engine could send an email at 9am and a LinkedIn message at 9:05am — two touchpoints consuming double attention capital with diminishing returns.

Rule A: V1.3 Section 7: Email and LinkedIn are separate channel sequences.

Rule B: V2 Section 7: Each touchpoint costs attention capital. V2 Section 1: EROA governs spend.

Engine Impact: Simultaneous multi-channel contact burns attention capital, creates perception of spam, and violates trust construction principles. The contact receives a wall of outreach in one morning.

Recommended Resolution: RESOLVE BY: Add CHANNEL-GAP-001: 'Minimum 48-hour gap between different channel touches to the same contact. If email sent today, LinkedIn message cannot be sent until day after tomorrow. Exception: if the contact engages on one channel (opens email, views LinkedIn), the other channel may be used 24 hours after engagement. Total weekly touchpoints per contact: max 3 across all channels.'

[HIGH] COH-E02: Western Send Window vs KSA Work Week

Sections: V1.3 Section 7 (Outreach Timing) vs V2 Section 17 (KSA Rules — preliminary)

Conflict Description: V1.3 Section 7 implies standard business hours for send windows (typical B2B: 8am-5pm). KSA's work week is Sunday-Thursday, not Monday-Friday. KSA business hours differ from Western norms. Ramadan further shifts timing. The outreach engine's timing module must be configured for KSA-specific windows, but V1.3's timing rules assume Western conventions.

Rule A: V1.3 Section 7: Implied Western business hours (Monday-Friday, 8am-5pm).

Rule B: KSA reality: Sunday-Thursday work week. Ramadan shifts. Different business hour norms.

Engine Impact: Emails sent on Friday (KSA weekend) or during prayer times get buried and never opened. The engagement metrics for KSA accounts appear artificially low, triggering incorrect scoring downgrades and tier demotions.

Recommended Resolution: RESOLVE BY: Add KSA-TIME-001: 'KSA send windows: Sunday-Thursday, 9am-12pm and 2pm-4pm (AST). No sending during Friday/Saturday. Ramadan: reduce daily volume by 50%, shift windows to 10am-12pm and 9pm-11pm. Prayer time blackouts: configurable per bank. Weekend definition is locale-specific, not hardcoded.'

[MEDIUM] COH-E03: Signal Polling Frequency: Undefined System Parameter

Sections: V1.3 Section 3 (Signal Sources) vs V2 Section 15 (Resilience) vs V2 Section 16 (Temporal Intelligence)

Conflict Description: V1.3 Section 3 specifies polling Google News, SAMA RSS, LinkedIn, etc. but does not specify how often. V2 Section 15 (Resilience) implies rate limiting and backoff. V2 Section 16 (Temporal Intelligence) implies time-sensitive signal detection. The polling frequency determines the engine's responsiveness: too slow misses time-sensitive signals, too fast burns API credits and risks rate limiting by source APIs.

Rule A: V1.3 Section 3: Signal sources listed but no polling frequency specified.

Rule B: V2 Sections 15 + 16: Resilience (rate limits) and temporal intelligence (timeliness) pull in opposite directions.

Engine Impact: Without specified polling frequency, the engineer picks an arbitrary interval. Too slow (daily) and time-sensitive signals arrive stale. Too fast (every minute) and API costs explode.

Recommended Resolution: RESOLVE BY: Add SIG-POLL-001: 'Polling frequency by source type: SAMA RSS (every 2 hours), Google News (every 4 hours), LinkedIn (every 6 hours), Hiring platforms (daily). Emergency mode: if a CRITICAL signal is detected, all sources poll immediately (one-time burst). Rate limit backoff: exponential with 2x multiplier, max 24-hour delay per source.'

[MEDIUM] COH-E04: Ramadan Blackout vs Signal Freshness Decay

Sections: V2 Section 17 (KSA Rules) vs V2 Section 4 (EPIS) vs V2 Section 16 (Temporal Intelligence)

Conflict Description: Ramadan is approximately 30 days. If the engine implements a Ramadan outreach blackout, signals collected during Ramadan will decay by 30+ days before they can be acted upon. By the time Ramadan ends, signals that were fresh and high-priority at collection are stale and low-scored. The engine effectively loses an entire month of intelligence.

Rule A: V2 Section 17: Ramadan blackout on outreach execution.

Rule B: V2 Section 4 + 16: Signals decay over time. 30-day-old signals score near zero.

Engine Impact: Post-Ramadan, all accounts score low because all signals are stale. The engine interprets this as 'no opportunities' when in fact opportunities were merely frozen during Ramadan.

Recommended Resolution: RESOLVE BY: Add RAMADAN-FRESH-001: 'During Ramadan blackout, signal freshness decay is paused (frozen at collection-time value). Post-Ramadan, freshness decay resumes from the frozen value, not from the calendar date. Signals collected during Ramadan get a 30-day freshness bonus to compensate for the forced delay. Enrichment continues during Ramadan; only outreach execution is blacked out.'

CATEGORY F: INTEGRATION AND DATA FLOW CONFLICTS

[CRITICAL] COH-F01: PostgreSQL vs HubSpot: Source of Truth Conflict

Sections: V1.3 Section 11 (Technology Stack) vs V1.3 Section 12 (CRM Sync) vs V2 Section 12 (Distributed Reality)

Conflict Description: V1.3 Section 11 specifies PostgreSQL as the database and HubSpot as the CRM. V1.3 Section 12 specifies POST /sync-hubspot for synchronization. But no rule specifies which system is the source of truth for each data type. If a sales rep updates an account's tier in HubSpot manually and the scoring engine computes a different tier in PostgreSQL, which wins? V2 Section 12 (Distributed Reality Coordination) implies PostgreSQL for engine-managed data, but the handoff boundary is unclear.

Rule A: V1.3 Section 12: CRM sync is bidirectional (implied by 'synchronize').

Rule B: V2 Section 12: Engine is the source of truth for scored data (implied but not explicit).

Engine Impact: Bidirectional sync without source-of-truth rules causes data oscillation: PostgreSQL writes tier HOT, HubSpot overwrites to WARM (sales rep change), next sync PostgreSQL overwrites back to HOT. The sales rep's input is lost; the scoring engine's input is delayed.

Recommended Resolution: RESOLVE BY: Add SOT-001: 'PostgreSQL is source of truth for: scores, tiers, engagement metrics, signal data, AI-generated content, workflow state. HubSpot is source of truth for: deal stage, deal amount, close date, owner assignment, manual notes. Sync direction per field is explicit. Conflict resolution: last-writer-wins with human override flag (if sales rep manually changed a field, that field is locked from engine overwrite for 72 hours).'

[HIGH] COH-F02: Enrichment Sources Without EPIS Confidence Scores

Sections: V1.3 Section 5 (Enrichment Sources) vs V2 Section 4 (EPIS Reality Confidence Model)

Conflict Description: V1.3 Section 5 lists enrichment sources: Apollo, Clearbit, Clay, LinkedIn. V2 Section 4 establishes the EPIS (Evidence, Proximity, Independence, Specificity) Reality Confidence Model as the governing spine for all integrations. But no EPIS scores are assigned to enrichment sources. Is Apollo data higher confidence than Clearbit? Is LinkedIn self-reported data (Proximity: high, Independence: low) treated the same as Clearbit firmographic data (Proximity: low, Independence: high)?

Rule A: V1.3 Section 5: Enrichment sources listed without confidence weighting.

Rule B: V2 Section 4: EPIS model requires confidence scoring for all data inputs.

Engine Impact: Without EPIS scores on enrichment data, the engine treats all enrichment equally. A self-reported LinkedIn title (low independence, high proximity) carries the same weight as a verified Clearbit firmographic record (high independence, low proximity).

Recommended Resolution: RESOLVE BY: Add EPIS-ENRICH-001: 'Each enrichment source receives a base EPIS profile. Apollo: E=0.7 P=0.6 I=0.7 S=0.8. Clearbit: E=0.8 P=0.5 I=0.9 S=0.7. LinkedIn: E=0.6 P=0.9 I=0.3 S=0.6. Clay: inherits EPIS from upstream source. These profiles modify the confidence of enriched data when it enters the scoring engine.'

[HIGH] COH-F03: Signal Immutability vs Enrichment Rescoring

Sections: V1.3 Section 19.4 (Signals Table) vs V2 Section 4 (EPIS) vs V2 Section 10 (Reality Calibration)

Conflict Description: V1.3 Section 19.4 states signals should NEVER be overwritten — all signals remain historically queryable. V2 Section 10 (Reality Calibration) implies that signal impact scores may be recalibrated as the engine learns. V2 Section 4 (EPIS) applies time decay to confidence. If signals are immutable but their impact scores change via recalibration, are we modifying the signal or creating a derived score?

Rule A: V1.3 Section 19.4: Signals are immutable. Never overwritten.

Rule B: V2 Section 10: Learning loops recalibrate signal impact scores.

Engine Impact: If the engineer treats recalibrated scores as signal modifications, immutability is violated. If they create separate derived-score records, the data model needs a new entity that does not exist.

Recommended Resolution: RESOLVE BY: Add SIG-IMMUTE-001: 'Raw signals are immutable (V1.3 Section 19.4 governs). Derived scores (impact, EPIS confidence, freshness) are computed values stored in a separate Signal_Score_History table, keyed by signal_id + computation_timestamp. Recalibration creates new derived score records; it never modifies raw signal records. Historical queries can reconstruct any past scoring state by joining signals with their contemporary derived scores.'

[MEDIUM] COH-F04: Two Email Execution Tools: No Selection Criteria

Sections: V1.3 Section 11 (Technology Stack) vs V1.3 Section 7 (Outreach)

Conflict Description: V1.3 Section 11 lists both Instantly and Smartlead as email execution tools. No rule specifies when to use which, or whether they serve different functions (e.g., cold outreach vs warm nurture), or whether one is primary and the other is failover.

Rule A: V1.3 Section 11: Email Execution = Instantly / Smartlead (listed as alternatives).

Rule B: V1.3 Section 7: Outreach execution references 'email tool' generically.

Engine Impact: The engineer picks one arbitrarily or implements both without routing logic. If both are active simultaneously, the same contact could receive emails from both tools, violating channel coordination rules and potentially triggering spam filters.

Recommended Resolution: RESOLVE BY: Add EMAIL-TOOL-001: 'Primary email tool: Instantly (for automated sequences). Smartlead: standby/failover. Add a tool selector in the orchestration config. Only one email tool is active per account at any time. Tool switching requires sequence completion or manual override. Contact email-tool mapping is stored to prevent dual-tool sends.'

CATEGORY G: GOVERNANCE AND LEARNING LOOP CONFLICTS

[CRITICAL] COH-G01: Constitutional Constraint vs Automatic Weight Update

Sections: V2 Section 13 (Learning Governance) vs V1.3 Section 10.3 (Analytics Workflow) vs V2 Section 1 (Meta-Principles)

Conflict Description: V2 Section 13 establishes that Section 1 Meta-Principles are constitutional — no learning loop can override them. V1.3 Section 10.3 says the analytics engine should 'Update Scoring Weights' automatically. V2 Section 1 defines specific scoring dimension weights as part of the meta-principle layer. If scoring weights are meta-principles, the analytics engine cannot update them automatically. If they are operational rules, the constitutional constraint does not apply.

Rule A: V2 Section 13: Meta-Principles are non-negotiable. No learning loop overrides.

Rule B: V1.3 Section 10.3: Analytics engine updates scoring weights automatically.

Engine Impact: Either the learning loop violates constitutional constraints (by auto-updating weights that are meta-principles), or it is unnecessarily restricted (if weights are operational, not constitutional).

Recommended Resolution: RESOLVE BY: Clarify the hierarchy explicitly. Add CONST-WEIGHT-001: 'Scoring DIMENSIONS (Signal, Regulatory, Reachability, Warmth) are meta-principles — no learning loop may remove a dimension. Scoring WEIGHTS (35%, 30%, 20%, 15%) are operational rules — learning loops may adjust weights within floor/ceiling constraints (10%-50%) per WEIGHT-FLOOR-001. The distinction: what we measure is constitutional; how much weight each measurement carries is operational.'

[HIGH] COH-G02: Small-Sample Recalibration vs Anti-Goodhart Safeguard

Sections: V2 Section 13 (Learning Governance — Anti-Goodhart) vs V1.3 Section 10 (Analytics) vs V2 Section 10 (Reality Calibration)

Conflict Description: V2 Section 13 establishes anti-Goodhart safeguards: the system must not optimize for metrics in ways that degrade the underlying objective. V1.3 Section 10 says to analyze performance and refine scoring. But with a small initial portfolio (10-20 accounts in MVP), recalibration from such a small sample is statistically unreliable and maximally susceptible to Goodhart effects — one lucky outcome could skew all future scoring.

Rule A: V2 Section 13: Anti-Goodhart safeguards required in all learning loops.

Rule B: V1.3 Section 10: Analyze performance and refine. No minimum sample size specified.

Engine Impact: A single successful outreach to one account could cause the engine to recalibrate all weights toward that account's profile, over-fitting to N=1.

Recommended Resolution: RESOLVE BY: Add GOODHART-SAMPLE-001: 'No automatic weight recalibration until minimum 50 scored engagements across at least 15 distinct accounts. Pre-threshold period uses A/B testing with control group (10% of outreach uses randomized weights) to build the statistical foundation. Post-threshold recalibration uses Bayesian updating with strong priors (initial weights) that requires substantial evidence to shift significantly.'

[MEDIUM] COH-G03: Prompt Versioning vs AI Model Registry: Redundant Governance

Sections: V1.3 Section 15.6 (Prompt Pitfalls) vs V2 Section 2 (Entity 27: AI_Model_Registry) vs V1.3 Section 19.6

Conflict Description: V1.3 Section 15.6 says to store prompts in database with version tracking. V1.3 Section 19.6 describes the AI Generations Table storing prompts, outputs, model used, QC results. V2 Section 2 defines Entity 27 (AI_Model_Registry) which catalogs AI models with versions and capabilities. But prompt versions and model versions are tracked in different entities with no explicit link. A prompt that works with one model may fail with another.

Rule A: V1.3 Section 19.6: Prompts stored with model_used per generation.

Rule B: V2 Entity 27: Models tracked separately with their own versioning.

Engine Impact: When a model is upgraded, all prompts must be re-tested. But there is no mechanism to flag which prompts need re-testing when a model version changes.

Recommended Resolution: RESOLVE BY: Add PROMPT-MODEL-001: 'Each prompt version is tagged with validated model versions. When a model is updated in AI_Model_Registry, all prompts validated only against the old version are flagged for re-testing. Re-testing uses the QC Agent with 10 representative inputs. Link AI_Generations to both prompt_version_id AND model_version_id.'

CATEGORY H: KSA-SPECIFIC AND COMPLIANCE CONFLICTS

[HIGH] COH-H01: PDPL Compliance vs Enrichment Data Collection Scope

Sections: V1.3 Section 25.3 (Compliance) vs V1.3 Section 5 (Enrichment) vs V2 Section 4 (EPIS) vs V2 Section 17

Conflict Description: V1.3 Section 25.3 says the platform should align with PDPL requirements. V1.3 Section 5 specifies enriching contacts with emails, hiring trends, LinkedIn activity, firmographic data. V2 Section 17 will cover KSA-specific rules including PDPL. But the enrichment engine's data collection scope is defined WITHOUT reference to PDPL. Saudi Arabia's Personal Data Protection Law restricts processing of personal data without legitimate basis.

Rule A: V1.3 Section 5: Enrichment collects emails, hiring data, LinkedIn activity.

Rule B: V1.3 Section 25.3: Align with PDPL requirements (no specific data mapping).

Engine Impact: The enrichment engine may collect data that PDPL prohibits without consent. Compliance failure in a Saudi regulatory environment is an existential business risk, not a technical bug.

Recommended Resolution: RESOLVE BY: Add PDPL-ENRICH-001: 'Each enrichment data type must be mapped to a PDPL legal basis before collection. Legitimate interest applies to publicly available business data (company news, public filings, LinkedIn public profiles). Personal contact data (email, phone) requires opt-in or legitimate interest documentation per PDPL Article 5. Enrichment pipeline adds a legal_basis_code to each data record. Records without a legal basis are flagged and require compliance review before use.'

[HIGH] COH-H02: Arabic Outreach vs English-Only AI Architecture

Sections: V1.3 Section 7 (AI Architecture) vs V2 Section 7 (Trust Construction) vs KSA Market Reality

Conflict Description: V1.3 Section 7 specifies AI message generation using Claude and GPT models. All prompt engineering examples and QC rubrics are in English. KSA banking executives may prefer Arabic communication. V2 Section 7 (Trust Construction) emphasizes cultural resonance and relationship building. But the AI architecture has no Arabic language capability specification.

Rule A: V1.3 Section 7: AI generation is English-only (all examples, prompts, QC in English).

Rule B: KSA market: Arabic is the primary business language for many senior banking executives.

Engine Impact: English-only outreach to Arabic-preferring executives fails the trust construction principle. The engine cannot personalize by language preference without an Arabic capability.

Recommended Resolution: RESOLVE BY: Add LANG-KSA-001: 'Language preference is stored per Contact and per Account. Default for KSA: Arabic for initial outreach, English for technical content. AI generation pipeline adds a language selector. Arabic outputs require separate QC rubric and potentially different AI model or fine-tuning. MVP: English-only with Arabic as P2 feature. P2: Arabic generation with native-speaker QC review.'

[MEDIUM] COH-H03: Vision 2030 References vs Hallucination Rules

Sections: V1.3 Section 16.3 (AI Principles) vs V2 Section 6 (Knowledge Acquisition) vs KSA Context

Conflict Description: V1.3 Section 16.3 prohibits hallucination and requires grounded context only. V2 Section 6 (Knowledge Acquisition) treats each target bank as a living causal model. KSA outreach often references Vision 2030, Saudi National Bank strategy, SAMA regulatory initiatives, etc. The AI may need to reference these in outreach. But the specificity matters: generic Vision 2030 references are always safe; specific claims about a bank's Vision 2030 alignment require evidence.

Rule A: V1.3 Section 16.3: Grounded context only. No hallucination.

Rule B: KSA market context: Vision 2030 references expected but specificity varies.

Engine Impact: The AI either avoids Vision 2030 references entirely (missing cultural context) or makes specific claims about a bank's alignment without evidence (hallucination violation).

Recommended Resolution: RESOLVE BY: Add V2030-REF-001: 'Two levels of Vision 2030 reference: (1) MARKET-LEVEL (always allowed): general statements about KSA digital transformation, fintech growth, open banking trends. No bank-specific claims. (2) BANK-LEVEL (evidence-required): reference specific bank initiatives only when grounded in a signal with EPIS confidence ≥ 0.7 (annual report, press release, executive speech). Label in the prompt chain which level is used.'

MASTER RESOLUTION PRIORITY TABLE

Not all conflicts are equal. The following priority order reflects engine-collapse risk, with the highest-risk conflicts first. This is the order in which resolutions must be confirmed before implementation can proceed.

#	Conflict ID	Severity	Category	Core Issue
1	COH-A01	CRITICAL	Anchor	Account vs Opportunity design - foundational
2	COH-B01	CRITICAL	Scoring	Two competing scoring models - gates everything
3	COH-A02	CRITICAL	Anchor	Score belongs to Account or Opportunity
4	COH-C01	CRITICAL	State	HOT state vs HOT tier name collision
5	COH-A03	CRITICAL	Anchor	Tier assignment for multi-Opportunity Accounts
6	COH-D01	CRITICAL	Autonomy	MVP review vs Autonomy Ladder - governs all AI
7	COH-C02	CRITICAL	State	Mid-outreach tier downgrade - no transition defined
8	COH-F01	CRITICAL	Integration	PostgreSQL vs HubSpot source of truth
9	COH-G01	CRITICAL	Governance	Constitutional constraint vs auto weight update
10	COH-B02	HIGH	Scoring	Hysteresis band vs weight recalibration
11	COH-B03	HIGH	Scoring	Three different freshness models
12	COH-C03	HIGH	State	Reply triggers handoff unconditionally vs intent check
13	COH-E01	HIGH	Timing	Email and LinkedIn channel collision
14	COH-E02	HIGH	Timing	Western send window vs KSA work week
15	COH-D02	HIGH	Autonomy	QC temp vs generation temp retry loop
16	COH-D03	HIGH	Autonomy	n8n zero logic vs workflow decision nodes
17	COH-A04	HIGH	Anchor	CRM Opportunity creation timing
18	COH-A05	HIGH	Anchor	Engagement events entity attribution
19	COH-F02	HIGH	Integration	Enrichment sources without EPIS scores
20	COH-F03	HIGH	Integration	Signal immutability vs enrichment rescoring
21	COH-B04	HIGH	Scoring	Scoring frequency and debounce
22	COH-C04	HIGH	State	DORMANT re-activation path missing
23	COH-G02	HIGH	Governance	Small-sample recalibration anti-Goodhart
24	COH-H01	HIGH	Compliance	PDPL vs enrichment data collection
25	COH-H02	HIGH	Compliance	Arabic outreach vs English-only AI architecture
26	COH-B05	MEDIUM	Scoring	Static signal priority vs contextual significance
27	COH-B06	MEDIUM	Scoring	Weight floor vs learning loop freedom
28	COH-C05	MEDIUM	State	Account state OPPORTUNITY vs Entity Opportunity
29	COH-D04	MEDIUM	Autonomy	Hallucination prohibition vs hypothesis generation
30	COH-E03	MEDIUM	Timing	Signal polling frequency undefined
31	COH-E04	MEDIUM	Timing	Ramadan blackout vs signal freshness
32	COH-F04	MEDIUM	Integration	Two email tools no selection criteria
33	COH-G03	MEDIUM	Governance	Prompt versioning vs model registry gap
34	COH-H03	MEDIUM	Compliance	Vision 2030 references vs hallucination rules

NEW RULES INTRODUCED BY RESOLUTIONS

Each conflict resolution introduces new rule IDs that must be integrated into the main document. This is the master index of all new rules proposed by this audit.

Rule ID	Source Conflict	Summary
ANCHOR-001	COH-A01	Account = relationship container. Opportunity = decision unit.
TIER-MULTI-001	COH-A03	Account tier = MAX(Opportunity tiers). Routing uses Opportunity tier.
CRM-OPP-001	COH-A04	Internal E38 and HubSpot Deal are separate lifecycle objects.
ENG-OPP-001	COH-A05	Engagement events carry opportunity_id from triggering message.
SCORE-DUAL-001	COH-B01	Dual-score: Account Attractiveness (0-100) + Effective Opportunity.
HYST-RECAL-001	COH-B02	Hysteresis widens during weight recalibration transition period.
FRESH-UNIFY-001	COH-B03	Single freshness model: Section 16 temporal curves are authoritative.
SCORE-DEBOUNCE-001	COH-B04	5-minute coalescing window for scoring. CRITICAL signals override.
SIG-CONTEXT-001	COH-B05	Signal priority = base x contextual modifier. MVP: base only.
WEIGHT-FLOOR-001	COH-B06	Dimension weights floor 10%, ceiling 50%. >5pp change = human approval.
TIER-NAME-001	COH-C01	Tiers renamed TIER_1/2/3. No state name may duplicate a tier name.
STATE-TIER-MID-001	COH-C02	Tier downgrade mid-sequence: pause, re-evaluate, transition or terminate.
REPLY-INTENT-001	COH-C03	Reply intent classification: POSITIVE/NEUTRAL/NEGATIVE/OOO/UNSUBSCRIBE.
DORMANT-REACT-001	COH-C04	DORMANT is not terminal. Re-activation on new signal, review, or override.
AUTO-MVP-001	COH-D01	All MVP operations cap at A1. Promotion requires 100 decisions + governance.
QC-TEMP-001	COH-D02	QC uses rubric-based evaluation. Max 3 regenerations, then human escalation.
ORCH-BOUNDARY-001	COH-D03	n8n routes events. FastAPI makes decisions. Boundary defined.
HYPOTHESIS-001	COH-D04	Internal hypothesis (allowed) vs output hallucination (prohibited).
CHANNEL-GAP-001	COH-E01	48-hour gap between channels. Max 3 weekly touches per contact.
KSA-TIME-001	COH-E02	KSA send windows: Sun-Thu 9-12, 2-4 AST. Ramadan adjustment specified.
SIG-POLL-001	COH-E03	Polling frequency per source type. Emergency burst on CRITICAL signal.
RAMADAN-FRESH-001	COH-E04	Freshness decay frozen during Ramadan. 30-day bonus post-Ramadan.
SOT-001	COH-F01	Source of truth per field. PostgreSQL for engine data, HubSpot for deal data.
EPIS-ENRICH-001	COH-F02	Each enrichment source receives a base EPIS profile.
SIG-IMMUTE-001	COH-F03	Raw signals immutable. Derived scores in Signal_Score_History table.
EMAIL-TOOL-001	COH-F04	Primary: Instantly. Smartlead: failover. One tool active per account.
CONST-WEIGHT-001	COH-G01	Dimensions are constitutional. Weights are operational (adjustable).
GOODHART-SAMPLE-001	COH-G02	Min 50 engagements across 15 accounts before auto-recalibration.
PROMPT-MODEL-001	COH-G03	Prompt versions tagged with validated model versions.
PDPL-ENRICH-001	COH-H01	Each enrichment data type mapped to PDPL legal basis before collection.
LANG-KSA-001	COH-H02	Language preference per Contact. MVP: English. P2: Arabic generation.
V2030-REF-001	COH-H03	Market-level references always allowed. Bank-level requires EPIS >= 0.7.

SECTION CROSS-REFERENCE MATRIX

This matrix shows which sections are involved in conflicts, helping engineers understand the blast radius of each resolution. A change to resolve COH-A01, for example, requires updating Sections 1, 2, 4, 5, 7, 8, 9, 12, 19, and 23.

Section	Conflicts Involved	Count
Section 1 (Meta-Principles)	A01, A02, A03, B01, B06, D04, G01	7
Section 2 (Entity Model)	A01, A05, F03, G03	4
Section 3 (Modules)	D03	1
Section 4 (Scoring)	A02, B01, B02, B04, B05, B06	6
Section 5 (Tiers/Enrichment)	A03, B01, C01, C02	4
Section 6 (Knowledge Acq.)	D04, H03	2
Section 7 (Outreach/Trust)	C02, C03, E01, E02, H02	5
Section 8 (Engagement/Trust Pres.)	A05, C02, C04	3
Section 9 (Handoff)	A04, C03	2
Section 10 (Analytics/Reality Cal.)	B02, B06, F03, G01, G02	5
Section 11 (Decision Partnership)	D02	1
Section 12 (CRM/Dist. Reality)	A04, F01	2
Section 13 (Learning Governance)	B06, G01, G02	3
Section 15 (Resilience)	B04, D01, E03	3
Section 16 (Temporal Intelligence)	B03, C04, E03, E04	4
Section 17 (KSA Rules)	E02, E04, H01, H02, H03	5
Section 19 (DB Schema)	A01, B03, F03, G03	4
Section 21 (n8n Architecture)	D03	1
Section 23 (State Machine)	A01, C01, C04, C05	4

RECOMMENDED RESOLUTION PATH

WHAT THIS AUDIT ESTABLISHES

This audit identifies 34 cross-section conflicts across 8 categories. 11 of these are **CRITICAL** — meaning they will cause the engine to produce incorrect behavior, deadlock, or data corruption if left unresolved.

The three load-bearing conflict clusters are:

1. The Anchor Model (COH-A01 through A05) — The Account-vs-Opportunity tension runs through every section. Until this is resolved with concrete dual-entity specifications (Account for relationships, Opportunity for deals), every downstream section inherits the ambiguity.

2. The Scoring Duality (COH-B01) — Two scoring models (V1.3 composite vs V2 Effective Opportunity) with different scales, semantics, and entity anchors. Until the dual-score architecture is explicitly specified, Section 5 (tiers), Section 6 (enrichment triggers), and Section 7 (outreach decisions) cannot be coherently implemented.

3. The State-Tier Confusion (COH-C01, C02) — The name collision between state HOT and tier HOT combined with missing state transitions for mid-lifecycle tier changes will cause the most visible operational failures (duplicate outreach, dropped sequences, inconsistent CRM records).

Recommended approach: Resolve **CRITICAL** conflicts in the order listed in the Master Resolution Priority table. Each resolution produces new rule IDs that must be cross-referenced into all affected sections. **HIGH** conflicts should be resolved before production launch. **MEDIUM** conflicts can be resolved during engineering review.

RESOLUTION WORKFLOW

Step 1: Confirm or modify each **CRITICAL** conflict resolution (9 conflicts, Priority 1-9).

Step 2: For each confirmed resolution, assign the new rule ID to the appropriate section(s) using the Section Cross-Reference Matrix.

Step 3: Update the master rule index in each affected section.

Step 4: Resolve **HIGH** conflicts (14 conflicts, Priority 10-25) before production launch.

Step 5: Resolve **MEDIUM** conflicts (9 conflicts, Priority 26-34) during engineering review.

Step 6: Re-run coherence audit after resolutions are integrated to verify no new conflicts were introduced by the resolutions themselves (resolution-induced conflicts).

Estimated Resolution Effort:

CRITICAL: 2-3 governance sessions to confirm resolutions + 1 week engineering integration.

HIGH: 1-2 governance sessions + 2 weeks engineering integration.

MEDIUM: Can be resolved inline during engineering review.

AUDIT LIMITATIONS

This audit cross-references the V1.3 FSD base document and V2 deep-specification layers as captured in project knowledge and conversation history. The full rule-level content of the 20+ PDF artifacts produced in prior sessions is reconstructed from conversation summaries and code fragments — not from reading the actual PDF files (which must be uploaded per session for authoritative access).

Specific rule IDs may have evolved between sessions. The audit identifies conflict PATTERNS that hold regardless of specific rule numbering.

V2 Sections 17-19 have not received deep-specification passes. Additional conflicts may surface as those sections are built. The KSA-specific conflicts (Category H) are preliminary — V2 Section 17's full specification will likely surface additional tensions.

Resolution recommendations are architectural proposals, not confirmed decisions. Each must be reviewed and confirmed through the governance process before implementation.